

K. J. Somaiya School of Engineering, Mumbai-77

Department of Sciences and Humanities

Course Name:	Basic of Electrical Engineering Laboratory	Semester:	I
Course Code:	316U06L102		
Date of Performance:	21/09/2026	Batch No:	1
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Faculty Sign&Date:		Grade/Marks:	/ 20

Experiment No: 3

Title: Thevenin' s & Nortons Theorem

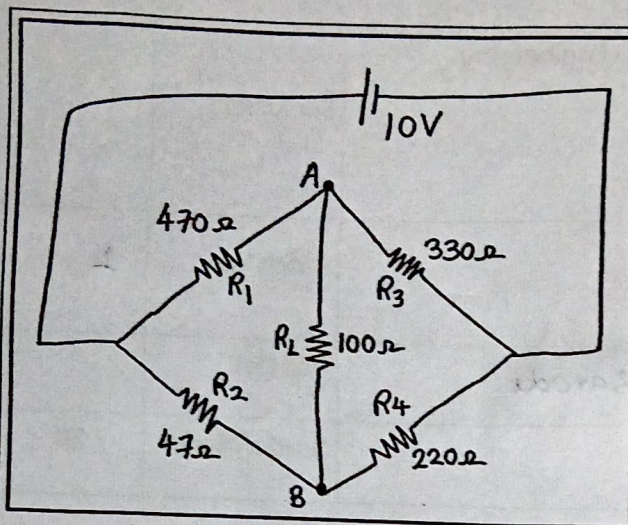
Aim and Objective of the Experiment:

- To Verify for Thevenin' s & Norton' s Theorem for the given circuit

COs to be achieved:

CO1: Analyze resistive DC networks using various network theorems.

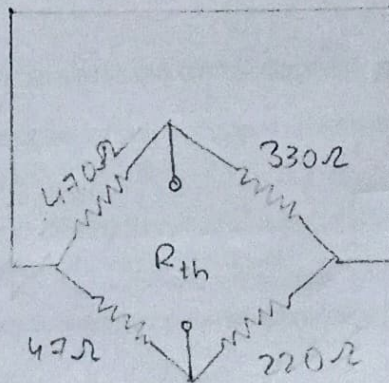
Circuit Diagram for Thevenins & Nortons



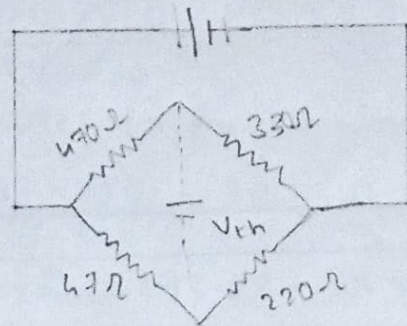
1. THEVENIN' S THEOREM

Solve for Thevenin' s

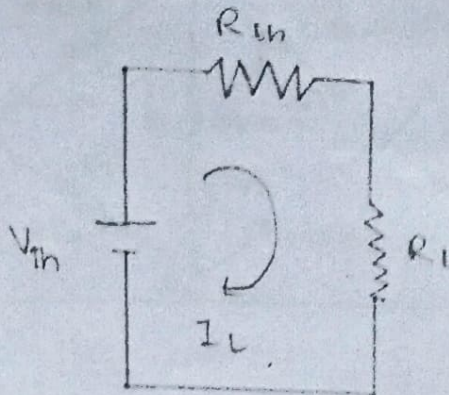
Task 1: Circuit Diagram to measure R_{TH} :



Task 2: Circuit Diagram to measure V_{TH} :



Task 3: Thevenin's Equivalent circuit



Stepwise-Procedure:

Thevenin's Theorem:

1. Connect the circuit as shown in the circuit diagram.
2. Set 10V and measure open circuit voltage V_{Th} across load terminals A and B.
3. Replace all voltage sources by Short circuit and measure R_{Th} across terminals A and B as per the circuit diagram shown in the figure.
4. Draw Thevenin's equivalent circuit and determine the value of load current from it.
5. Verify the results theoretically.

Calculations:

For calculating R_{Th} ,

R_1 is parallel to R_3 ,

$$\therefore R_1 \parallel R_3 = \frac{470 \times 330}{470 + 330} = 193.88 \Omega ; \therefore R_{13} = 193.88 \Omega$$

R_2 is parallel to R_4 ,

$$\therefore R_2 \parallel R_4 = \frac{47 \times 220}{47 + 220} = 38.73 \Omega ; \therefore R_{24} = 38.73 \Omega$$

$$\therefore R_{Th} = R_{13} + R_{24} = 193.88 + 38.73 = 232.61 \Omega$$

For calculating V_{TH}

$$V_A = 10 \times \left(\frac{330}{470 + 330} \right) = 4.125V$$

$$V_B = 10 \times \left(\frac{220}{47 + 220} \right) = 8.23V$$

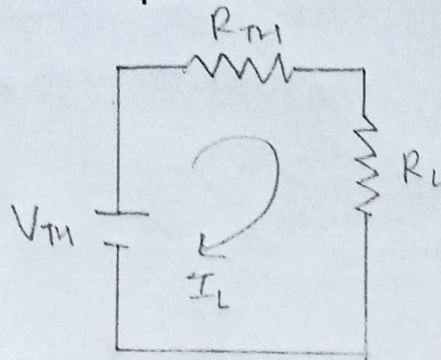
$$V_{TH} = V_A - V_B = 4.125 - 8.23 = -4.105 = 4.105V$$

$$\therefore I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{4.105}{232.6 + 100} = \frac{4.105}{332.6} = 12.3mA$$

Observation Table:-

	$V_{th}(V)$	$R_{th}(\Omega)$	$I_L (mA)$
Theoretical Value	4.1V	232.6 Ω	12.3mA
Practical Value	3.6V	228 Ω	12mA

Draw Thevenin' s Equivalent circuit:-



Thevenins Theorem: (Using VLAB)

Using following link open VLAB experiment for Thevenin' s Theorem Verification.

<http://vlabs.iitkgp.ernet.in/asnm/exp3/index.html>

1. Verify Thevenin' s Theorem using VLAB.
2. Follow the procedure as mentioned in VLAB.
3. Draw Thevenin' s equivalent circuit and determine the value of load current.
4. Verify the results theoretically.

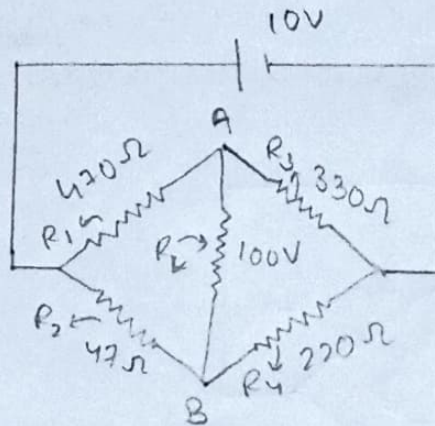
Attach the screenshots of the circuit, results

Observation Table:

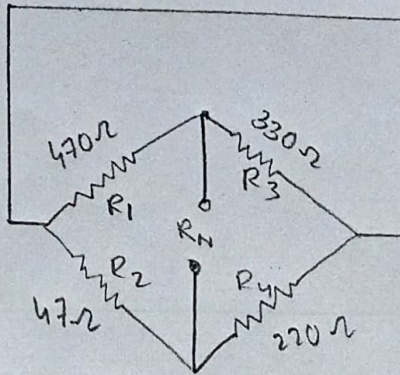
Serial no. of Observation	Load Current(I_L) from case 1	Load Voltage(V_L)	Load Resistance (R_L)= V_L/I_L	Thevenin Voltage(V_{th}) from case 2(a)	2nd Voltage source(v) for case 2(b)	Ammeter Reading(I) from case 2(b)	Thevenin Resistance $R_{th}=V/I$	Load current (I_L)= $V_{th}/(R_{th}+R_L)$
1st	0.10645	53.225	500	82.500	220	0.80000	275.00	0.10645
2nd	0.11613	58.065	500	90.000	215	0.78182	275.00	0.11613
3rd	0.11137	56.7987	510	89.302	215	0.73665	291.86	0.11137
4th	0.12094	58.0512	480	91.176	225	0.82152	273.88	0.12094
5th	0.11023	57.31959995	520	89.286	250	0.86207	290.00	0.11023

NORTONS THEOREM:-

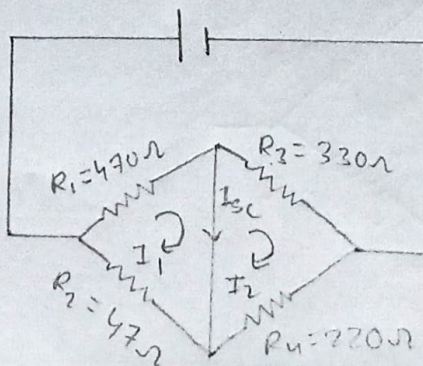
Circuit Diagram to Verify Norton Theorem for the Circuit



Task 1: Circuit Diagram to measure R_N :



Task 2: Circuit Diagram to measure I_N (Isc):



Norton's Theorem: (Using VLAB)

Using following link open VLAB experiment for Norton's Theorem Verification.

<http://vlabs.iitkgp.ernet.in/asnm/exp2/index.html#>

5. Verify Norton's Theorem using VLAB.
6. Follow the procedure as mentioned in VLAB.
7. Draw Norton's equivalent circuit and determine the value of load current.
8. Verify the results theoretically
9. Attach the screenshots of the circuit, results

Calculations:

For Norton's theorem,

$$R_N = R_{TH} = 232.61 \Omega$$

For calculating $I_N (I_{sc})$

$$I_N = \frac{V_{TH}}{R_N} = \frac{4.1}{232.61} = 17.62 \text{ mA}$$

For calculating I_L ,

$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

$$= \frac{0.01762 \times 232.61}{232.61 + 100}$$

$$\therefore I_L = 12.32 \text{ mA}$$

Screenshots of Norton's Theorem using VLAB

Observation Table:

Serial no. of Observation	Load Current (I_L) from case 1	Load Voltage (V_L)	Load Resistance (R_L) $\rightarrow V_L/I_L$	Norton current (I_{sc}) from case 2(a)	2nd Voltage source (V) from case 2(b)	Ammeter Reading (I) from case 2(b)	Norton Resistance $R_N = V/I$	Load current (I_L) $= I_{sc} \cdot R_N / (R_N + R_L)$
1st	0.28387	85.161	300	0.67692	110	0.50769	216.67	0.28387
2nd	0.26907	86.102	320	0.71707	110	0.57236	192.19	0.26907
3rd	0.26710	93.485	350	0.81606	180	1.0570	170.29	0.26709
4th	0.17772	57.226	322	0.49502	117	0.64874	180.35	0.17772
5th	0.14800	52.096	352	0.33775	150	0.54636	274.54	0.14800

Observation Table:-

	Rth/RN (Ω)	I_N (mA)	I_L (mA)
Theoretical Value	232.6 Ω	17.62mA	12.32mA
Practical Value			

Draw Norton' s Equivalent circuit

Post Lab Questions:

1. State what is mean by linear network.

→ A linear network is an electrical circuit whose components - such as ideal resistors, capacitors and inductors - have constant values regardless of the current flowing through them or the voltage across them.

The fundamental defining characteristic of a linear network is that it satisfies the principle of superposition.

2. State the different applications of linear network.

→ Linear networks are essential for circuit analysis using superposition. Thevenin's and Norton's theorems. They are widely used in signal processing and audio equipment for filtering and noise reduction, power distribution systems for load modeling and control systems to maintain predictable, proportional system responses without distortion.

Conclusion:

Thevenin's & Norton's theorem verified from

For the given circuit.

Signature of faculty in-charge with Date: